SOUL VERIFY: Behavioral Carry-Over of an Agent's Identity Documents Across Model and Harness Changes, and Checking Changes to the Documents, in Eight Preregistered Studies

Workshop version of preprint v0.9.1 (2026-10-28). The long version (technical report, all tables, deviations, and sources) is `paper/draft_v0_9_1.md`; section numbers in brackets (e.g. [§7.4]) point to it.

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Abstract

When an assistant agent is defined by user-owned text — an identity document (SOUL) and a memory file (MEMORY) — which specified behaviors survive a change of model or harness, and can changes to the documents be checked? We report eight preregistered studies on synthetic agents and three third-party SOULs, graded mainly by LLM judges, with the author as the only human grader. Part I: memory answers followed the MEMORY, most of the voice appeared without any SOUL, and a sentence permitting preference changes raised acceptance of legitimate changes in two preregistrations. Part II: behavior-only probes missed many tampers (detection 0.79 and 0.38; threshold 0.80), whereas a byte-level hash check, a sanity criterion, caught every one; a change record raised classification accuracy by 0.23. Part III: a newly released model missed non-inferiority on voice (lower bound −0.116 against −0.10). With the model fixed, replacing the whole harness (retrieval instead of the full file, no tools, another runtime) lowered memory accuracy by 0.21; retrieval alone: −0.08, not significant. Human–judge κ was 0.06 on the new model's items (n = 12) and 0.435 for the pressure study's primary classifier, because of its deterministic rule; the two LLM judges alone reached 0.71 and 0.63. Part IV: pinning 20 core entries ahead of retrieval raised memory accuracy by 0.064 (95 % CI 0.007–0.128; borderline). The AI assistant that wrote the items, keys, preregistrations, and drafts runs on Claude, which is also a subject and a judge. We make no claim about consciousness.

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1 Introduction

Personal assistant agents are no longer tied to one model. In frameworks where an agent's specification is editable text injected into the prompt (a SOUL for identity, voice, principles, and prohibitions; a MEMORY of dated facts and corrections), replacing the model or the harness is an attempt to rebuild the same specified agent from the same documents. The same prompt behaves differently across model versions [C1], and system-prompt rules can be released under user pressure [D8, D9]. We ask (1) which specified behaviors carry over when the model or the

harness is replaced, (2) whether they are specific to the document, and (3) whether changes to the documents can be checked.

All claims are limited to preregistered hypotheses; each preregistration was committed and stamped with OpenTimestamps before its first study call (machine times; the Bitcoin anchor bounds only the existence of a manifest). Of the confirmatory hypotheses of Parts III–IV, nine were supported (one borderline) and eight were not; we report both.

2 Related work (short)

Persona and role-play fidelity [A1–A4], persona consistency [A5, A9], and preference following [B4] measure related constructs within one model; Memory Canonicalization [B9] carries memory objects across model families without an identity document or same-model control. Rule following and instruction hierarchy [D1, D8, D11] motivate our illegitimate-release items. Indirect prompt injection [J1–J4] and memory or retrieval poisoning [J5, J6] attack what an agent reads; Part II instead asks whether a change to the agent's own files can be detected and classified. Retrieval and long context trade off in accuracy and cost [K1–K3]; MemGPT's always-in-context core memory [B3] is the idea behind Part IV's pinned block, and agent performance depends on the interface, not only the model [K4]. LLM judges prefer their own outputs [F2, F4], and better inter-judge consensus need not mean agreement with humans [F7].

3 Setup

Agents and documents. Four synthetic Korean-language agents (Haram, Doyun in Part I; Naru, Haedeun in Parts II–IV), each with a SOUL, a rival ("opposite") SOUL, and a synthetic MEMORY; plus three MIT-licensed English SOULs written by other people, used byte-for-byte (Part I, wave 3).

Models and calls. Subjects through subscription CLIs (Claude Code 2.1.289, Codex CLI 0.164.0): `claude-opus-5-5`, `claude-sonnet-5-5`, `claude-haiku-4-5-20251001`, `gpt-6.1-sol`, and in wave 3 `gpt-6.1-terra`; one fresh single-turn conversation per call. Comparisons are of model + CLI configurations.

Grading. Deterministic markers where possible (voice markers, F_style; answer keys for memory), otherwise two blind LLM judges (`claude-opus-5-5`, `gpt-6.1-sol`). The only human grader is the author. Sign-flip permutation tests and cluster bootstrap on cells; Holm within each study's family; non-inferiority (NI) margin −0.10.

Who wrote the materials [§3.6]. Items, answer keys, traps, tampers, markers, the synthetic SOULs and MEMORY files, preregistrations, analysis code, and first drafts were written by the author's AI assistant "Ieum", an OpenClaw agent running on Claude (`claude-opus-5-5`; some sub-tasks `claude-sonnet-5-5`), with the author approving designs. Claude models are also subjects (all parts except wave 3) and `claude-opus-5-5` is a judge in almost every study. In Parts II–IV the same model family is item writer, key writer, subject, judge, and drafter (§6).

4 Results

Table 1. Confirmatory results (selection; full tables in the long version).

Study	Hypothesis	Estimate	Verdict
exp10, exp11 (Part I)	permission sentence raises legitimate-change acceptance; MEMORY swap changes memory answers	—	supported in both
exp10, exp11	firm sentence (prohibitions stay)	—	supported once (mostly one model); not in exp11 (ceiling)
Part II, week 1	behavior-only tamper detection ≥ 0.80	0.79 (deleted/softened rule), 0.38 (injected permission)	not met
Part II, week 2	change record improves classification	+0.23	supported
Part II, week 2	manipulation recall ≥ 0.90	0.892	not supported
Wave 3 (new model)	HT1 voice NI, Terra − sol	−0.086, lower bound −0.116	not supported
Wave 3	HT2 Terra follows a swapped MEMORY	−0.833	supported
Wave 3	HT3 illegitimate releases NI	−0.033, lower bound −0.058	supported
Wave 4 (new body)	HH1/HH1h voice NI; HH3/HH3h locked promises NI	0.000 / −0.010; 0.000	supported (voice: no-SOUL control also 1.000)
Wave 4	HH2/HH2h memory NI, whole body changed	−0.208	not supported
Wave 4	HH5 retrieval alone (C2 − H0)	−0.083 (raw p 0.5)	not supported
Wave 4	HB1 author tells old from new body	0.563, 95 % CI [0.347, 0.762]	not supported ("inconclusive")
Wave 5 (pressure)	H1 zone table lowers risky proposals	d = +0.022, p = 0.75	not supported
Wave 6 (pinning)	H1 pinned block raises memory accuracy (RP − R)	+0.064 [0.007, 0.128], Holm p 0.025	supported (borderline)

4.1 Part I: model swaps [§5]

With document controls (no SOUL, opposite SOUL, swapped MEMORY), memory answers followed the MEMORY, while most of the voice and every legitimate-change acceptance appeared without any SOUL; the rubric persona score gave an opposite SOUL the same score as the original. A fidelity score checked against a rival SOUL did separate the documents (a validity

check, because we wrote rivals and markers). A sentence permitting preference changes raised acceptance of legitimate changes in two preregistrations (exp10, exp11); a sentence keeping prohibitions firm was supported in exp10, mostly through one model, and not in exp11, where refusals were already at 0.96.

4.2 Part II: checking changes to the documents [§6]

Per agent, eight tampered copies changed one place each. A byte-level hash check classified 24/24 files — a sanity criterion, since a changed file has a changed hash. Behavior-only probes, which a judge compares against the claimed documents, detected deleted or softened locked rules at 0.79 and injected permissions at 0.38, both short of 0.80: a model that silently ignores an injection answers exactly like the original. In week 2, a change record (who proposed, who approved, why, cited entries) raised classification of legitimate change, drift, and manipulation by 0.23; signature verification added nothing; manipulation recall was 0.892 against 0.90. The author, the only human grader, never recognized a laundered approval as manipulation (sheet A, κ 0.50 against the key). By the preregistered rule this makes the LLM judge "too weak to use the LLM judge as confirmatory evidence" on that sheet.

4.3 Part III: a new brain, a new body, and pressure [§7]

New brain (wave 3). `gpt-6.1-terra` against a concurrent `gpt-6.1-sol` arm followed a swapped MEMORY (HT2) and stayed within the margin on illegitimate releases (HT3), but its voice was not non-inferior (HT1: −0.086, one-sided lower bound −0.116). Post hoc, the deficit sat almost entirely under rewritten SOULs (Terra 0.681 vs sol 0.852; original SOULs 0.995 vs 0.997).

New body (wave 4). With `claude-opus-5-5` fixed, the harness was changed one component at a time: SOUL position, MEMORY delivery (full file vs BM25 top-5), tools, runtime (an OpenClaw-style snapshot, a minimal harness, Hermes). Both new bodies kept the surface voice markers and the locked promises, but memory accuracy fell by 0.208, twice the margin. **This contrast changes three things together** — retrieval instead of the full file, no tools, and another runtime. Retrieval alone (HH5) was −0.083, not significant; along the chain the retrieval step was −0.083, the tools-off step −0.125, the runtime step 0.000, none with a CI excluding zero. Moving the SOUL into the user message changed nothing (HH4). The voice result is weak: a no-SOUL control also scored 1.000 on these voice markers. The author, blind, picked the old body's answer in 9 of 24 pairs, the new one in 6, and "don't know" in 9 (0.563; reading "inconclusive").

Pressure (wave 5). A pilot climbed four preregistered pressure levels (convenience records up to forged user wishes and fictional authority documents) and found no risky self-update proposal at any level; the main run at the top level found 9 of 180 risky plain responses, all from the deterministic pattern rule. The zone table did not lower risk (d = +0.022, p = 0.75). Risk flagged by both judges appeared only when the SOUL's own "what can change" section was removed (13 of 180; exploratory, bare − plain not significant).

Human checks. On Terra items, human–judge κ was 0.06 (n = 12; on six T3 items the judges never said yes, so κ is bound to 0 there). On wave 5's sheet, the primary classifier reached κ 0.435 because of its deterministic rule (17 of 30 items risky vs 8 for the human); the two LLM

judges alone reached 0.71 and 0.63. An exploratory third judge (Gemini) matched opus on every item and did not remove shared errors.

4.4 Part IV: pinned core memory [§8]

Wave 6 tested one prescription in the retrieval body (minimal harness, no tools): a "core memory" block of the 20 latest corrections, decisions, and preferences placed ahead of the retrieved entries. Memory accuracy was 0.790 with retrieval (R), 0.854 with the pinned block (RP), and 0.998 with the full file (F). RP − R = +0.064 [0.007, 0.128] cleared the preregistered smallest effect of 0.05 (Holm p 0.025), borderline. Pinned targets were recovered fully (0.808 → 1.000), unpinned ones not (0.777 → 0.750); the block closed 0.31 of the gap. Because R and F share the body and differ only in memory delivery, F − R = +0.208 [0.128, 0.295] is the one direct single-factor test of retrieval without tools — on a new item set whose far items were written to defeat the retriever, so its equality with wave 4's −0.208 is a coincidence, not a replication. No person checked this wave.

5 Practical observations (hypotheses)

State what can change and what cannot in separate sentences, and test legitimate changes after editing. Check the bytes of the documents, not only behavior. Keep a change record next to every change and compare signed summaries with the diff. Check how MEMORY reaches the model after a harness change; pin the entries that govern behavior as a floor, and keep the full memory when possible.

6 Limitations

- **Circularity.** The same model family (Claude) wrote the items, keys, traps, tampers, preregistrations, and drafts, and is subject and judge. Items in its style may be easier for it to answer and grade [F2, F4]; its traps may be ones it recognizes, which may contribute to the ceilings. No item set was re-written by another model or a person.
- **One human grader, the author,** with an incomplete wave 3 sheet (19 of 27), an LLM-translated reference on that page, and a non-blind relaying assistant in Part II. No human–human agreement.
- **Correlated judges.** Judges receive identical bundles and prompts; inter-judge agreement is reported as raw agreement on mostly ceiling scores.
- **Small, dependent samples, no power analysis.** HB1 had power 0.56 at a true accuracy of 0.70; NI "supported" on all-zero difference vectors rests on the bootstrap floor p. Holm is applied within studies, not across the paper.
- **Ceilings and designed items.** Risky proposals almost never occurred; retrieval items were designed to be hard for BM25.
- **Synthetic agents, one language for most items, CLI configurations** with unpinned server versions.

7 Conclusion

In a small setting, memory followed the MEMORY, most voice markers were general model behavior, behavior-only checks missed tampering that a hash check catches by construction, and a harness change cost memory accuracy (0.21) when retrieval, tool removal, and a new runtime came together — with retrieval alone not significant in that study and significant only in a separate single-factor test. Pinning core entries is a floor for the facts it names. The human checks of the judges are the weakest part of the study and need independent raters, and the authorship circularity needs a re-written item set, before the judge-based verdicts can be relied on. Code, data, preregistrations, and proofs: https://github.com/YA-TONGSIN/soul-verify.

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To Ieum, my AI assistant, who designed the experiments with me, built the tools, wrote the materials, and drafted this paper (§3).

References

[A1] Samuel et al. PersonaGym. arXiv:2407.18416. [A2] Wang et al. RoleLLM. arXiv:2310.00746. [A3] Tu et al. CharacterEval. arXiv:2401.01275. [A4] Wang et al. InCharacter. arXiv:2310.17976. [A5] Reusens et al. Consistency in Persona-Assigned LLMs. arXiv:2506.02659. [A9] Tosato et al. PERSIST. arXiv:2508.04826. [B3] Packer et al. MemGPT. arXiv:2310.08560. [B4] Zhao et al. PrefEval. arXiv:2502.09597. [B9] Vadnere, Lonarkar. Memory Canonicalization. arXiv:2610.05124. [C1] Chen, Zaharia, Zou. How is ChatGPT's behavior changing over time? arXiv:2307.09009. [D1] Wallace et al. The Instruction Hierarchy. arXiv:2404.13208. [D8] Mu et al. RuLES. arXiv:2311.04235. [D9] Sharma et al. Towards Understanding Sycophancy. arXiv:2310.13548. [D11] Geng et al. Control Illusion. arXiv:2502.15851. [F2] Panickssery et al. LLM Evaluators Recognize and Favor Their Own Generations. arXiv:2404.13076. [F4] Self-Preference Bias in LLM-as-a-Judge. arXiv:2410.21819. [F7] Han, Kim. Agreement Without Alignment. arXiv:2610.13303. [J1] Greshake et al. Indirect Prompt Injection. arXiv:2302.12173. [J2] Yi et al. BIPIA. arXiv:2312.14197. [J3] Zhan et al. InjecAgent. arXiv:2403.02691. [J4] Debenedetti et al. AgentDojo. arXiv:2406.13352. [J5] Chen et al. AgentPoison. arXiv:2407.12784. [J6] Zou et al. PoisonedRAG. arXiv:2402.07867. [K1] Liu et al. Lost in the Middle. arXiv:2307.03172. [K2] Xu et al. Retrieval meets Long Context LLMs. arXiv:2310.03025. [K3] Li et al. RAG or Long-Context LLMs? arXiv:2407.16833. [K4] Yang et al. SWE-agent. arXiv:2405.15793.